

CBSE Practice Worksheet

Class 7 | Science | Difficulty: Hard | Set E

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. Which organelle is involved in programmed cell death?
(A) Lysosome
(B) Mitochondria
(C) Peroxisome
(D) Ribosome
2. Which reaction is endothermic?
(A) Combustion
(B) Neutralization
(C) Photosynthesis
(D) Respiration
3. Which principle explains the working of a hydraulic lift?
(A) Archimedes
(B) Pascal
(C) Bernoulli
(D) Newton
4. What is the escape velocity from Earth's surface?
(A) 8 km/s
(B) 11.2 km/s
(C) 15 km/s
(D) 25 km/s
5. What is the hybridization of carbon in methane?
(A) sp
(B) sp^2
(C) sp^3
(D) sp^3d
6. What is the wavelength of Lyman series limit?
(A) 91.2 nm
(B) 121.6 nm
(C) 656.3 nm
(D) 364.6 nm
7. Which type of mutation involves change in single nucleotide?
(A) Deletion
(B) Insertion
(C) Point mutation
(D) Frameshift
8. What is the efficiency of an ideal Carnot engine operating between 300K and 400K?
(A) 25%
(B) 33%
(C) 50%

(D) 75%

9. What is the coordination number in FCC structure?

- (A) 6
- (B) 8
- (C) 12
- (D) 14

10. What is the oxidation state of Mn in KMnO_4 ?

- (A) +2
- (B) +4
- (C) +6
- (D) +7

Section B: Fill in the Blanks (1 mark each)

- 11. The IUPAC name of $\text{CH}_3\text{CH}_2\text{CHO}$ is ____.
- 12. The enzyme that converts prothrombin to thrombin is ____.
- 13. The Krebs cycle occurs in the ____ of mitochondria.
- 14. In $\text{S}_\text{N}2$ reactions, the rate depends on both ____ and nucleophile.
- 15. Raoult's law applies to ____ solutions.
- 16. The process of splitting water during photosynthesis is called ____.
- 17. The band gap in semiconductors is typically ____ eV.
- 18. The magnetic field inside a solenoid is given by $B = \mu \blacksquare$ ____.

Section C: Short Answer Questions (2 marks each)

- 19. Describe the mechanism of nerve impulse transmission.
- 20. Derive the equation for kinetic energy of gases from kinetic theory.
- 21. Explain the structure and working of a nuclear reactor.
- 22. Explain the mechanism of DNA replication.
- 23. Explain Le Chatelier's principle with examples.
- 24. Describe the light-dependent and light-independent reactions of photosynthesis.
- 25. Derive the expression for electric field due to a dipole.

--- End of Worksheet ---

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